# 1. Block Diagram Representation

A **block diagram** is a pictorial representation of the functions performed by each component and of the flow of signals. It shows the relationship between input and output of every element through its transfer function.

## Elements of a block diagram

- **Block** – rectangle with the T.F. G(s) inside; output = G(s) × input. Arrow = direction of signal flow.
- **Summing point** – circle with ⊠ ; signs (+/−) show addition or subtraction of signals.
- **Take-off (branch) point** – dot from which the same signal goes to more than one place; value does not change.
- **Forward path** – from input to output; **feedback path** – from output back to the summing point.

**Open loop T.F. = $G(s)H(s)$ , Feed-forward T.F. = $C(s)/E(s) = G(s)$**

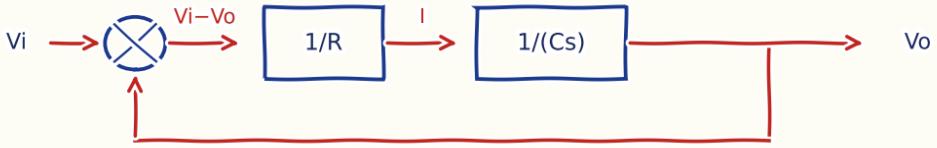
**Closed loop T.F. : $C(s)/R(s) = G(s)/[1 + G(s)H(s)]$ ; $E(s)/R(s) = 1/[1 + G(s)H(s)]$**

## Procedure to draw the block diagram of an electrical system

- Write the differential equations of the network (KVL / KCL).
- Take Laplace transform with zero initial conditions.
- Rearrange each equation as (output of block) = (T.F.) × (input), choosing one variable per block.
- Connect blocks using summing points and take-off points; check that every variable appears once as an output.

## Example: Series RC circuit (input $V_i$, output $V_o$ across C)

$$I(s) = [V_i(s) - V_o(s)] / R \; ; \; V_o(s) = I(s) / (Cs)$$




*Using the feedback formula (G = 1/(RCs), H = 1):*

$$V_o(s)/V_i(s) = [1/(RCs)] / [1 + 1/(RCs)] = 1 / (RCs + 1)$$

### Example: Series R–L–C circuit (output across C)

*Loop equations: $V_i - V_o = (R + Ls)\, I$ and $V_o = I/(Cs)$. So we use a block $1/(R+Ls)$ (error voltage → current), a block $1/(Cs)$ (current → $V_o$) and unity feedback:*

$$V_o/V_i = 1 / (LC\, s^2 + RC\, s + 1)$$

## 2. Block Diagram Algebra (Reduction Rules)

| Rule | Original | Equivalent |
|---|---|---|
| 1. Cascade blocks | $G1 \rightarrow G2$ | $G1 \cdot G2$ |
| 2. Parallel blocks | $G1 \pm G2$ (summed) | $G1 \pm G2$ |
| 3. Feedback loop (neg.) | G with feedback H | $G / (1 + GH)$ |
| 4. Shift summing point beyond a block | $R \rightarrow (+) \rightarrow G$, extra input X before G | X must be multiplied by G after block $(X \cdot G)$ |
| 5. Shift summing point ahead of a block | Extra input X after G | X must be divided by G (X/G) before block |
| 6. Shift take-off point beyond a block | Take-off before G | Take-off after G, multiply by 1/G in tapped branch |
| 7. Shift take-off point ahead of a block | Take-off after G | Take-off before G, multiply tapped branch by G |
| 8. Interchange summing points | $a \pm b \pm c$ | Order can be changed freely |
| 9. Eliminate feedback H | Feedback path H(s) | Unity feedback with $G/(1+G(H-1))$ approach: put 1/H before the loop and H after |
| 10. Move take-off across summing point | – | Not allowed directly; add a summing point on the branch |

**★ Golden rule of reduction: the product of transfer functions in the forward path must remain the same, and the product around any loop must remain the same.**

## Steps for reducing a complicated block diagram

- Combine all **cascade** blocks. Combine all **parallel** blocks.
- Eliminate all **inner feedback loops**.
- Shift summing points to the left and take-off points to the right of the loop.
- Repeat steps 1–3 until a single block remains.

### Worked example

*Forward path G1 → G2 → G3, inner feedback H1 around G2, outer feedback H2 from output to input summing point.*

*Step 1: inner loop = G2/(1+G2H1). Step 2: cascade = G1·[G2/(1+G2H1)]·G3. Step 3: outer loop:*

**C/R = G1G2G3 / ( 1 + G2H1 + G1G2G3H2 )**

## 3. Signal Flow Graph (SFG)

*An SFG is a graphical representation of a set of linear algebraic equations. It consists of **nodes** (variables) joined by directed **branches** (gains = transfer functions). It is simpler than block diagram reduction because Mason's gain formula gives the T.F. directly.*

### Terminology

| Term | Meaning |
|---|---|
| Node | Represents a system variable (sum of incoming signals) |
| Branch | Directed line segment between nodes with gain (transmittance) |
| Input (source) node | Only outgoing branches |
| Output (sink) node | Only incoming branches |
| Mixed node | Both incoming and outgoing branches |
| Path | Traversal of branches in the direction of arrows without repeating a node |
| Forward path | Path from input node to output node, no node crossed twice |
| Loop | Closed path starting and ending at the same node, no node crossed twice |
| Self loop | Loop with one branch and one node |
| Non-touching loops | Loops having no common node |
| Path gain / Loop gain | Product of branch gains along the path / loop |

### Properties of SFG

- Applicable to linear systems only; node value = sum of all incoming signals.
- Signal flows only in the direction of the arrow; a branch multiplies signal by its gain.
- Equations must be written as effect = $\Sigma$ (gain $\times$ cause).

### Converting a block diagram to SFG

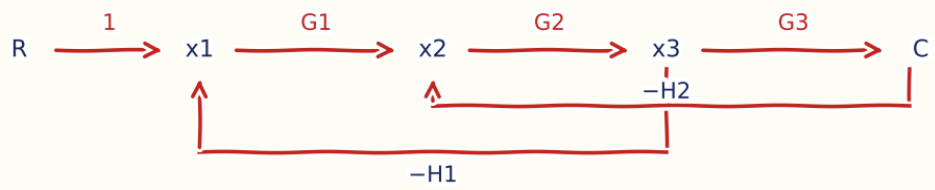
- Assign a node to the input, the output, every summing point and every take-off point.
- Draw a branch for each block with gain = T.F.; branch with gain $-1$ or $-H$ for negative feedback.
- Summing point node = sum of incoming branches (with their signs).

## 4. Mason's Gain Formula

$$T = C(s)/R(s) = ( \Sigma_k P_k \Delta_k ) / \Delta$$

- $P_k$ = gain of the $k^{th}$ forward path.
- $\Delta = 1 -$ (sum of all individual loop gains) + (sum of gain products of all possible pairs of non-touching loops) $-$ (sum of products of all possible triplets of non-touching loops) + ...
- $\Delta_k$ = value of $\Delta$ after removing all loops touching the $k^{th}$ forward path (put these loop gains to zero).

### Example 1




Forward path: $P_1$ = G1G2G3 (only one). Loops: $L_1$ = $-$G1G2H1 (x1→x2→x3→x1), $L_2$ = $-$G2G3H2 (x2→x3→C→x2). Both loops share node x2 → **touching**, so no

*non-touching products.*

$$\Delta = 1 - (L1 + L2) = 1 + G1G2H1 + G2G3H2 \; ; \Delta_1 = 1$$

$$T = G1G2G3 \; / \; ( 1 + G1G2H1 + G2G3H2 )$$

### Example 2 (with non-touching loops)

*Let a system have two forward paths $P_1 = G1G2G3$ and $P_2 = G4$, loops $L_1 = -G2H1$, $L_2 = -G3H2$ and $L_3 = -G1G2G3H3 - \ldots$; suppose $L_1$ and $L_2$ do not touch each other.*

$$\Delta = 1 - (L1 + L2 + L3) + (L1 \cdot L2)$$

**Path P1 touches all loops → $\Delta_1 = 1$ ; Path P2 touches none of L1, L2 → $\Delta_2 = 1 - L3$ (if L3 touches P2 then $\Delta_2 = 1 -$ (loops not touching P2))**

$$T = (P1 \cdot \Delta 1 + P2 \cdot \Delta 2) / \Delta$$

★ **Common mistakes: forgetting non-touching loop pair products, taking wrong sign of loop gains (negative feedback gives negative loop gain), and not recalculating Δk for each path.**

## 5. Compensation Techniques

*A **compensator** is an additional network/controller added to the system so that the performance specifications (steady-state error, peak overshoot, settling time, gain/phase margin) are satisfied when gain adjustment alone is not enough.*

| Type | Position | Used when |
|---|---|---|
| Series (cascade) compensation | In forward path, before the plant | Most common; easy design |
| Feedback (parallel) compensation | In a feedback path around part of the plant | Reduces effect of parameter variations |
| Feed-forward compensation | Parallel path from input | Improves disturbance rejection/speed |

## Lead compensator

*Passive RC lead network ($R_1$ in parallel with C, in series with $R_2$):*

$$G_c(s) = (1/\alpha)\,(1 + \alpha Ts)/(1 + Ts)\,,\ \alpha = (R1 + R2)/R2 > 1\,,\ T = R1R2C/(R1 + R2)$$

$$\text{Maximum phase lead: } \varphi_m = \sin^{-1}[(\alpha - 1)/(\alpha + 1)] \text{ at } \omega_m = 1/(T\sqrt{\alpha})$$

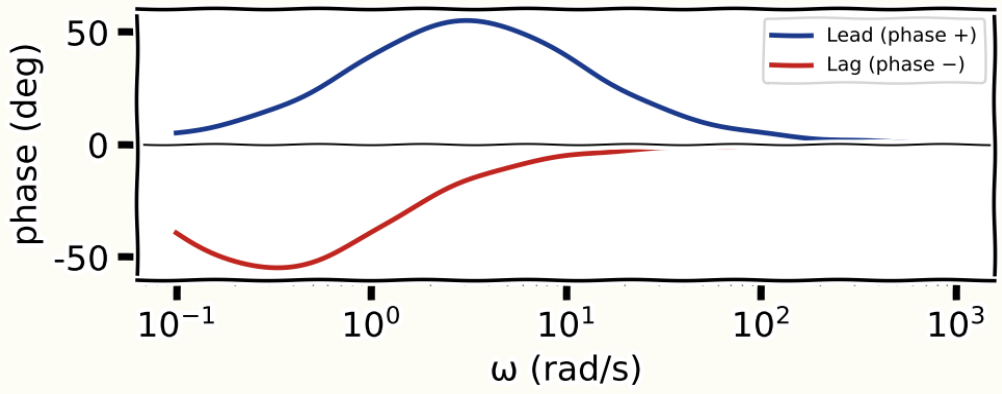
- *Zero lies closer to origin than pole (zero at $1/\alpha T$, pole at $1/T$). Acts like a high-pass filter / derivative action.*
- ***Effects:*** *adds positive phase → increases phase margin, damping and stability; increases bandwidth → faster response (lower rise & settling time); amplifies high-frequency noise; does not improve steady-state accuracy much.*
- *Used when the system needs better* ***transient response****.*

## Lag compensator

$$G_c(s) = (1 + Ts)/(1 + \beta Ts)\,,\ \beta = (R1 + R2)/R2 > 1\,,\ T = R2C$$

$$\text{Maximum phase lag: } \varphi_m = \sin^{-1}[(1 - \beta)/(1 + \beta)] \text{ at } \omega_m = 1/(T\sqrt{\beta})$$

- *Pole is closer to origin than the zero. Acts like a low-pass filter / integral action.*
- ***Effects:*** *reduces high-frequency gain (gain crossover moves to lower $\omega$); improves steady-state accuracy (increases error constants $K_p$, $K_v$ by $\beta$ factor); decreases bandwidth → slower response; reduces noise; stability margin gets better by lowering $\omega_{gc}$ but phase lag is added.*
- *Used when the system needs better* ***steady-state accuracy*** *with acceptable transient behaviour.*




## Lag–Lead compensator

$$G_c(s) = [\,(1 + T1\,s)(1 + T2\,s)\,] / [\,(1 + (T1/\gamma)\,s)(1 + \gamma\,T2\,s)\,],\ \gamma > 1$$

**(lead part: T1 s terms; lag part: T2 s terms)**

*Combines the advantages of both: lag part improves steady-state accuracy, lead part improves transient response and stability. Used when both are required.*

| Parameter | Lead | Lag | Lag–Lead |
|---|---|---|---|
| Pole/Zero | zero nearer origin | pole nearer origin | both |
| Phase | +ve | −ve (small at design ω) | −ve at low ω, +ve at high ω |
| Bandwidth | increases | decreases | increases |
| Rise/settling time | decreases | increases | decreases |
| Steady-state error | small change | reduces | reduces |
| Noise | amplified | attenuated | – |
| Like | PD controller | PI controller | PID controller |

# 6. Design of Compensators (Bode Plot Method)

## Lead compensator design steps

- Using the given static error constant ($K_v$ or $K_p$) determine gain K; plot Bode of KG(s).

- Find the existing phase margin (PM) and compute the extra phase needed: $\varphi_m = PM_{desired} - PM_{existing} + (5° \text{ to } 12° \text{ safety})$.
- Calculate $\alpha = (1 + \sin \varphi_m)/(1 - \sin \varphi_m)$.
- Find frequency where $|KG(j\omega)| = -10 \log \alpha$ dB; take it as new gain crossover $\omega_m$.
- $T = 1/(\omega_m \sqrt{\alpha})$; compensator = $(1 + \alpha Ts)/(1 + Ts)$; restore the gain loss ($K_c = \alpha$, if the network has $1/\alpha$ attenuation).
- Check PM, GM of the compensated system; repeat if specs are not met.

### Lag compensator design steps

- Set K from steady-state requirement; draw Bode of KG(s).
- Find the frequency $\omega_c$ at which phase = $-180° + PM_{desired} + (5° \text{ to } 12°)$ — new gain crossover.
- Measure gain of $KG(j\omega_c)$ in dB, set $20 \log \beta$ = that value → $\beta$.
- Place the zero one decade below $\omega_c$: $1/T = \omega_c/10$; pole at $1/(\beta T)$.
- $Gc(s) = (1 + Ts)/(1 + \beta Ts)$; verify the margins.

### Lag–lead design (outline)

Choose $\gamma$ from the required $K_v$ and the new gain crossover frequency; design lead part for phase margin; then the lag part ($T_2$) with zero at $\omega_c/10$; combine.

> ★ **Exam trick: $\alpha$ for lead is > 1 and $\beta$ for lag is > 1; always check that the phase of the lag network at $\omega_c$ is only a few degrees ($-5°$ to $-12°$).**

## 7. PID Controllers

The **PID controller** produces a control signal from the error e(t) by Proportional, Integral and Derivative actions:

$$u(t) = K_p\, e(t) + K_i \int e\, dt + K_d\, de/dt$$

$$G_c(s) = K_p + K_i/s + K_d\, s = K_p[\,1 + 1/(T_i\, s) + T_d\, s\,],\ K_i = K_p/T_i,\ K_d = K_p T_d$$

| Action | T.F. | Effect | Drawback |
|---|---|---|---|
| P | $K_p$ | Reduces rise time & steady-state error (not zero) | Offset remains, larger $K_p$ → oscillation |
| I | $K_i/s$ | Eliminates steady-state error (raises type by 1) | Slower, increases overshoot, may destabilise |
| D | $K_d s$ | Adds damping, reduces overshoot, anticipates error | Amplifies noise, no effect on steady-state error |
| PI | $K_p + K_i/s$ | Zero steady-state error to step; acts like lag | Slower transient |
| PD | $K_p + K_d s$ | Faster transient; acts like lead | High-frequency noise |

## Effect of increasing each gain (step response)

| Gain ↑ | Rise time | Overshoot | Settling time | Steady-state error |
|---|---|---|---|---|
| $K_p$ | Decrease | Increase | Small change | Decrease |
| $K_i$ | Decrease | Increase | Increase | Eliminate |
| $K_d$ | Small change | Decrease | Decrease | No change |

## Ziegler–Nichols tuning (ultimate-gain method)

Set I and D to zero, increase $K_p$ until sustained oscillation: ultimate gain $K_u$ and period $P_u$.

| Controller | $K_p$ | $T_i$ | $T_d$ |
|---|---|---|---|
| P | $0.5\,K_u$ | $\infty$ | 0 |
| PI | $0.45\,K_u$ | $P_u/1.2$ | 0 |
| PID | $0.6\,K_u$ | $P_u/2$ | $P_u/8$ |

# 8. More Solved Problems (exam type)

## Problem 1 – Mason's formula with a feed-forward path

*Nodes: R → E (gain 1), E → X (G1), X → C (G2), C → E (−H), and a direct branch R → C (G3).*

**Forward paths: P1 = G1G2 , P2 = G3**

**Loop: L1 = −G1G2H ⇒ Δ = 1 + G1G2H**

**P2 shares node C with L1 → Δ1 = Δ2 = 1**

**T = C/R = ( G1G2 + G3 ) / ( 1 + G1G2H )**

***Check by algebra:*** *C = G2G1E + G3R and E = R − HC ⇒ C(1 + G1G2H) = (G1G2 + G3) R ☒.*

## Problem 2 – Two non-touching loops

*Chain R →(G1)→ x1 →(G2)→ x2 →(G3)→ x3 →(G4)→ x4 →(G5)→ C. Feedbacks: −H1 from x2 to x1, −H2 from x4 to x3, −H3 from x4 to x2.*

**L1 = −G2H1 , L2 = −G4H2 , L3 = −G3G4H3**

**L1 and L2 do not touch (no common node) → product L1L2 = G2G4H1H2**

**Δ = 1 + G2H1 + G4H2 + G3G4H3 + G2G4H1H2**

**P1 = G1G2G3G4G5 touches all loops → Δ1 = 1**

**T = G1G2G3G4G5 / ( 1 + G2H1 + G4H2 + G3G4H3 + G2G4H1H2 )**

## Problem 3 – Lead compensator design (numerical)

*Plant G(s) = 4/[s(s+2)]. Specifications: $K_v = 20\ s^{-1}$, PM ≥ 50°.*

- *$K_v$ = lim sKG = 2K = 20 ⇒ K = 10, so KG(s) = 40/[s(s+2)].*
- *Gain crossover: $40/(\omega\sqrt{(\omega^2+4)}) = 1 \Rightarrow \omega_{gc} \approx 6.17$ rad/s; PM = 90° − $\tan^{-1}(6.17/2) \approx 18°$.*
- *Required lead: $\varphi_m = 50° - 18° + 5° = 37°$.*
- *$\alpha = (1 + \sin 37°)/(1 - \sin 37°) \approx 4.0$ ; extra gain at $\omega_m$ = 10 log α ≈ 6 dB.*

- Find $\omega$ where $|KG| = -6$ dB $\Rightarrow \omega_m \approx 8.84$ rad/s. $T = 1/(\omega_m\sqrt{\alpha}) \approx 0.0564$, $\alpha T \approx 0.227$.

**$G_c(s) = (1 + 0.227s)/(1 + 0.0564s)$ (with gain K = 10 retained)**
**New PM $\approx 90° - \tan^{-1}(8.84/2) + 37° \approx 50°$ ☒ ; $K_v$ unchanged = 20**

## Lag–lead RC network

**$E_o/E_i = (1 + R1C1\,s)(1 + R2C2\,s) / [\,(1 + R1C1\,s)(1 + R2C2\,s) + R1C2\,s\,]$**

Poles and zeros are real; the lag part acts at low frequencies and lead part at higher frequencies; at $\omega = 1/\sqrt{(R1C1R2C2)}$ the network phase is zero.

## Block diagram vs Signal flow graph

| Point | Block diagram | SFG |
|---|---|---|
| Representation | Blocks, summing & take-off points | Nodes and branches |
| Reduction | Step by step algebra (slow for big systems) | Mason's gain formula (direct) |
| Applicable to | Linear and non-linear (with blocks) | Linear systems only |
| Best for | Physical understanding of system | Quick overall T.F. |

**★ Remember: for the final answer always write the closed-loop T.F. in the form N(s)/D(s) and mention the characteristic equation D(s) = 0.**

## Important Questions

- State and explain the block diagram reduction rules with an example. (10M)
- Find C/R using Mason's gain formula for the given signal flow graph. (10M)
- Explain lead, lag and lag–lead compensators with circuits and T.F. (10M)
- Design a lead/lag compensator for the given specifications using Bode plot. (10M)
- Explain PID controller; effects of P, I, D actions; Z–N tuning. (5M/10M)